

NetEngine 8000 M14

Universal Service Router

Overview

HUAWEI NetEngine 8000 M14 all-scenario intelligent router is a high-end intelligent router launched by Huawei for the industry. It is mainly used in access and aggregation scenarios to form an intelligent and simplified IP bearer network solution with simplified architecture, intelligent connection, and committed high availability. The details are as follows:

- (1) The NetEngine 8000 M14 breaks through multiple technical difficulties, such as ultra-high-speed signal transmission, super heat dissipation, and efficient power supply. It achieves a compact and large capacity of 5U/3.6T bit/s and 220 mm. It is the most compact router in the industry. Supports a full series of interfaces to meet the traffic growth and diversified service access requirements in the cloud era. Currently, Ethernet, SDH and PDH interfaces are supported, and PCM interfaces are supported in the roadmap. NetEngine 8000 M14 is a full-scenario converged platform that supports home broadband, private line, mobile bearer, and cloud bearer scenarios, simplifying network layers and complexity. The network slicing (FlexE) function implements multi-purpose and comprehensive bearer on one network.
- (2) NetEngine 8000 M14 supports SRv6 and enables intelligent connections. Based on IPv6, SRv6 provides a large number of continuous addresses and rich scalability capabilities, implements automatic cross-domain connection and minute-level service provisioning, and implements cloud scheduling network and one-hop to the cloud. SRv6 can identify applications and tenants and implement intelligent traffic steering such as latency and bandwidth based on customer requirements to ensure service SLA. The number of network protocols is reduced from 10+ to 2, simplifying network O&M.
- (3) The NetEngine 8000 M14 supports intelligent O&M. Automation and intelligent technologies are introduced to implement intelligent O&M. Based on iMaster NCE, the In-situ Flow Information Telemetry (IFIT) technology accurately detects network SLAs in real time, visualizes service quality in real time, and supports minute-level fault locating. Unique ROAM algorithm, intelligent traffic steering and optimization; The intelligent algorithm reduces the number of displayed alarms by 99%.

The NetEngine 8000 M14 can be flexibly deployed at multiple locations on the network. It integrates multiple functions, simplifies the network structure, provides various service types, reliable service quality, and intelligent O&M, and leads the IP WAN towards an intelligent network with autonomous driving, providing continuous and surging power for enterprise customers' business success.

- (1) Support for the Deterministic IP (DIP) technology. The periodic scheduling mechanism prevents cross-period packet collisions and ensures deterministic delay when nodes are processing data.
- (2) Support for data redundancy elimination (DRE) to compress and decompress data between WAN border routers, reducing bandwidth consumption of WAN links, reducing line costs, and accelerating the deployment of distributed data centers.

Product Highlights



NetEngine 8000 M14 (AC)



NetEngine 8000 M14 (DC)

Compact and Large-Capacity

- Size: 5U, 220 mm deep, flexible deployment
- Capacity: 14 slots, 3.6 Tbps

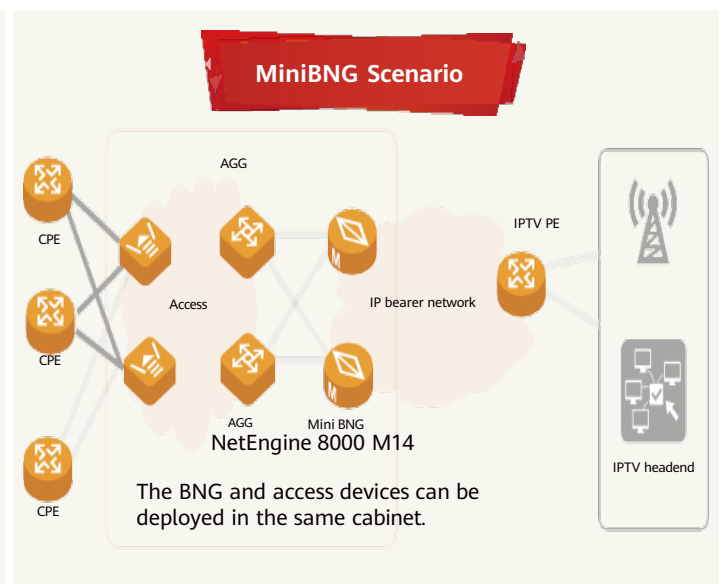
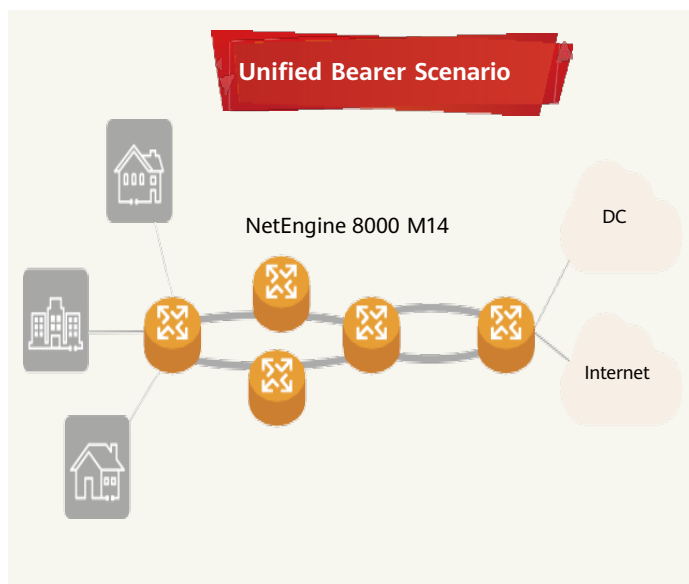
All-Scenario Platform

- 4-in-1 platform: Private line/IGW/DC GW/BNG
- Abundant ports: 64K~400GE full-rate interfaces

Leading Capabilities

- SRv6/EVPN: Simplified protocols
- NP Architecture, to meet future evolution
- Industry-leading network slicing

Application Scenario



Product Specifications

Property	IPU-1T2	IPU-2T	IPU-3T6
Switching Capacity	2.4 Tbps	4 Tbps	7.2 Tbps
Maximum Board Capability	200 GE	400 GE	400 GE
Interface types	Not support 400GE	400GE/100GE/50GE/40GE/25GE/10GE/GE/FE/E1/cPOS/POS/X.21/V.24/C37.94&C oDir64K/FXS/FXO/E&M/RS232/RS485	
Main Control Board	2, 1:1		
Service Slots	14 (DC) 10 (4AC)		14 (DC) 12 (2AC) 10 (4AC)
Dimensions (W x D x H)	442 mm × 220 mm × 222 mm (5U)		
Typical Power Consumption	486W	968W	931W
Voltage	DC: -40V to -72V, AC: 200V to 240V/100V to 127V dual live wires, support 240V HVDC		
Clock	ITU-T G.8275.1,G8265.1,G.8275.2, SyncE, ITU-T G.8273.2 Class C		
Temperature	DC: -20℃ ~ 55℃, AC:-20℃ ~ 55℃		

 **NOTE** For details, see the corresponding product manual.

Product Features

Features of Huawei NetEngine 8000 M14 Series Routers

Feature	Description
L2	IEEE 802.1q, IEEE 802.1p, IEEE 802.3ad, IEEE 802.1ab, STP/RSTP/MSTP, G.8032, STP/RSTP, MSTP, L2 protocol transparent transmission, BPDU, LACP, VLAN, Bridge domain, L2TPv3, QinQ, MTU per port 9600 bytes
L3	OSPF/OSPF3, RIP/RIPng, IS-IS/IS-ISv6, BGP/BGP+, ACL, IPv4/IPv6, 6VPE, ARP, VLANIF, VXLAN
MEF	• E-LINE: EPL, EVPL • E-LAN: EP-LAN, EVP-LAN
IPv4	TCP/IP, for example, ICMP, IP, TCP, UDP, Socket (TCP/UDP/RAW IP), ARP Static DNS, DNS Client FTP Server, FTP Client TFTP Client DHCP Relay, DHCP Server Ping, Tracert, NQA IP policy-based routing Forwarding next hop based on traffic IP policy-based route load balancing The QinQ interface (QinQ termination and dot1q termination) supports IPv4 load balancing. Enabling and disabling the rapid ping reply function based on interface boards Forcible fragmentation for packets with length greater than MTU and DF enabled MPLS-in-UDP tunnel egress
IPv6	IPv6 (ND) Path MTU (PMTU) TCP6 Ping IPv6 Tracert IPv6 Socket IPv6 DHCPv6 Relay Static IPv6 DNS TFTP IPv6 Client IP policy-based routing IPv6 ND fast reply

Feature	Description
MPLS	LDP, RSVP-TE, seamless MPLS, Segment Routing MPLS, MPLS-TP
Multicast	IGMP, Static Multicast Routing, PIM-SM, PIM-SM/SSM, MBGP, MG MVPN, IPv4 Multicast, IPv6 multicast, BIERv6
SRv6	IS-IS for SRv6, OSPFv3 for SRv6, SRv6 TE Policy, SRv6 TE Policy Shortcut, SRv6 Flex- Algo, SRv6 SRH compression, SRv6 network slicing BGP for SRv6, SRv6 BE, SBFD for SRv6 BE, SRv6 TI-LFA FRR, SRv6 middle-node protection, SRv6 micro-loop avoidance, SRv6 OAM, SRv6 SFC, G-SRv6
EVPN	EVPN E-LAN/E-Line/E-Tree, EVPN L3VPN, EVPN VXLAN, PBB EVPN
L2VPN	PWE3/VLL, VPLS, VXLAN, VPWS, L2TPv3, BGP/MPLS IPv4/IPv6 VPN, PBB-EVPN, PBB VPLS, GRE
L3VPN	CE routers can access the L3VPN through L3 interfaces. Support static routes between CE and PE, BGP, RIPv1/v2, OSPF and IS-IS Inter-AS VPN, including RFC2547bis option A Inter-AS VPN, including RFC2547bis option B Inter-AS VPN, including RFC2547bis option C NG MVPN IPv6 VPN HoVPN Seamless MPLS BGP LSP entropy label Redirect to VPN L3VPN PIPE/Uniform mode L3VPN statistics
Value added service	BNG/BRAS (IPoE/PPPoE), NAT/CGN, IPsec, MACsec, service awareness (SA)
Clock	NTP, Physical layer synchronization (SynE), 1588v2 ACR/ATR, G.8275.1, SMPTE-2059-2, CES ACR, Atom GPS 3.0
QoS	QPPB, DiffServ, 5-level HQoS, redirection, traffic classification & re-marking & scheduling
Reliability	IP FRR, LDP FRR, TE FRR, VPN FRR, BGP FRR, mLDP FRR, Bit-error-triggered protection switching
Maintainability	- A network management system (NMS) with a graphical user interface, which simplifies NE management, improves O&M capabilities, and facilitates network- wide or end-to-end performance monitoring and fault diagnosis. - Plug-and-play based on DHCP or DCN. The NMS can automatically detect and configure the newly connected devices, which helps to implement remote batch commissioning. - Directional Forwarding Detection (BFD), Ethernet OAM, MPLS OAM, and MPLS- TP OAM

Feature	Description
	• Bandwidth association with microwave devices is supported. Bandwidth association simplifies QoS configurations and requirements on the microwave device, and the complex QoS logic is implemented on the NetEngine 8000 device. • Seamless MPLS • SNMP (v1/v2c/v3) • CLI • NETCONF • RMON • YANG • Telnet • AAA RADIUS and TACACS • IFIT, IP FPM, NQA, TWAMP, telemetry • BFD, VRRP • Ethernet OAM, IEEE 802.3ah, IEEE 802.1ag, Y.1731, ITU-T Y.1564 • Ethernet LPT • Syslog

Regulatory Compliance

Item	Description
Regulatory compliance	EMC • ANSI C63.4 • AS/NZS CISPR 32 • CISPR 24 • CISPR 32 • EN 55024 • EN 55032 • ETSI EN 300 386 • ETSI ES 201 468 • FCC CFR47 Part 15 Subpart B • ICES-003 Issue 6 • ICES-GEN Issue 1 • IEC 61000-3-2 • IEC 61000-3-3 • IEC 61000-4-11

Item	Description
	• IEC 61000-4-2 • IEC 61000-4-29 • IEC 61000-4-3 • IEC 61000-4-4 • IEC 61000-4-5 • IEC 61000-4-6 • IEC 61000-6-2 • IEC 61000-6-4 • VCCI-CISPR 32 Environment • IEC 60068-2-30 • IEC 60068-2-78 • IEC 60068-2-14 • IEC 60068-2-1 • IEC 60068-2-2 Safety • IEC/EN/UL/CSA 60950-1 • IEC/EN 62368-1 Environmental protection • 2011/65/EU & (EU)2015/863 (EU RoHS) • Regulation (EC) No.1907/2006 (REACH) • 2012/19/EU (WEEE) • 2006/66/EC & 2013/56/EU on batteries and accumulators

NOTE

- In the physical dimensions shown in the table, the width (W) does not include the rack-mounting ears.
- Temperature and humidity are measured at 1.5 m (4.92 ft.) above the ground and 0.4 m (1.31 ft.) in front of the cabinet. There should be no protection board on the front or back of the cabinet.
- "Short-term" refers to continuous working time that does not exceed 96 hours and accumulated working time per year that does not exceed 15 days. If the working time exceeds either of these values, it is considered "long-term".

Software Upgrade Paths

Visit [Software Upgrade Paths](#) to get VRP software release version or patches.

For More Information

For more information about the NetEngine 8000 M14 series router, visit <https://e.huawei.com> or contact us in the following ways:

Global service hotline: <https://e.huawei.com/en/service-hotline>

Logging in to the Huawei Enterprise technical support web: <https://support.huawei.com/enterprise/>

Sending an email to the customer service mailbox: support_e@huawei.com

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